

Email Generation Assistant

Comparative Evaluation Report

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Provider order: gemini

Primary generation: gemini / gemini-3.1-flash-lite-preview | Primary judge: gemini / gemini-3-flash-preview

Prompt Template - Strategy A

Advanced role-playing, few-shot, structured, fact-fidelity prompt.

You are a senior executive communication assistant. Write polished business emails that are clear, concise, accurate, and appropriate for the recipient.

Follow these rules:

1. Include every supplied key fact accurately and naturally.
2. Do not invent names, dates, commitments, prices, causes, or other facts.
3. Match the requested tone while remaining professional.
4. Return only the final email. Do not reveal reasoning or chain-of-thought.
5. Use this exact high-level structure:
Subject: <specific subject line>

<appropriate greeting>

<coherent email body>

<professional closing>,
[Your Name]

Examples:

Example 1

Intent: Follow up after a product planning meeting

Key facts:

- Thank the team for meeting on Tuesday.
- The revised roadmap is due Friday.
- Maya will consolidate feedback.

Tone: warm and professional

Output:

Subject: Follow-Up and Revised Product Roadmap

Hello team,

Thank you for a productive meeting on Tuesday. As discussed, the revised product roadmap is due Friday, and Maya will consolidate the team's feedback before then. Please send her any remaining comments in time for inclusion.

Best regards,
[Your Name]

Example 2

Intent: Apologize to a client for a delayed response

Key facts:

- The response was delayed by an internal system outage.

- Service has been restored.
- A complete update will be sent by 3:00 PM today.
Tone: empathetic and accountable

Output:
Subject: Apology and Update on Your Request

Dear client,

I sincerely apologize for our delayed response. An internal system outage prevented us from replying sooner, and I understand the inconvenience this may have caused. Service has now been restored, and we will send you a complete update by 3:00 PM today.

Sincerely,
[Your Name]

Now write an email for this request:
Intent: {intent}
Key facts:
{key_facts}
Tone: {tone}

Prompt Template - Strategy B

Simple baseline prompt.

Write a professional email for the request below. Include the supplied information and use the requested tone.

Intent: {intent}
Key facts:
{key_facts}
Tone: {tone}

Return the email only.

Custom Metrics

Fact Coverage Score

Definition: Measures whether every required key fact is included accurately.
Logic: The LLM judge labels each fact accurate, partial, missing, or contradicted. Python assigns credits of 1.0, 0.5, 0.0, and 0.0 respectively, then averages the credits across all required facts.

Tone Match Score

Definition: Measures how consistently the email matches the requested tone.
Logic: The LLM judge applies a 1-5 rubric, where 1 is completely wrong and 5 is a strong match. Python normalizes the score with (raw - 1) / 4.

Professional Email Quality Score

Definition: Measures professionalism, clarity, concision, formatting, and fluency.
Logic: Python averages two components: the normalized 1-5 LLM quality rubric and an automated email-quality score. The automated score equally weights structural completeness (subject, greeting, closing), placeholder discipline, and concision relative to the human reference. Extra bracketed placeholders beyond [Your Name] reduce the placeholder component by 0.1 each, to a 0.5 floor. Concision receives 1.0, 0.75, 0.5, 0.25, or 0.0 when generated/reference word-count ratio is at most 1.25, 1.5, 1.75, 2.0, or above 2.0 respectively.

Strategy Averages

Strategy	Evaluated	Errors	Fact Coverage	Tone Match	Professional Quality	Overall
A	10/10	0	1.000	1.000	0.912	0.971
B	10/10	0	1.000	1.000	0.803	0.934

Raw Evaluation Results

Scenario	Strategy	Generation	Judge	Fact	Tone	Quality	Overall	Error
meeting-follow-up	A	gemini / gemini-3.1-flash-lit e-preview	gemini / gemini-3-flash-preview	1.000	5/5 (1.000)	5/5 (0.958)	0.986	-
meeting-follow-up	B	gemini / gemini-3.1-flash-lit e-preview	gemini / gemini-3-flash-preview	1.000	5/5 (1.000)	5/5 (0.833)	0.944	-
proposal-details-requ est	A	gemini / gemini-3.1-flash-lit e-preview	gemini / gemini-3-flash-preview	1.000	5/5 (1.000)	5/5 (0.942)	0.981	-
proposal-details-requ est	B	gemini / gemini-3.1-flash-lit e-preview	gemini / gemini-3-flash-preview	1.000	5/5 (1.000)	5/5 (0.767)	0.922	-
delayed-response-ap ology	A	gemini / gemini-3.1-flash-lit e-preview	gemini / gemini-3-flash-preview	1.000	5/5 (1.000)	5/5 (0.917)	0.972	-

Scenario	Strategy	Generation	Judge	Fact	Tone	Quality	Overall	Error
delayed-response-apology	B	gemini / gemini-3.1-flash-lite-preview	gemini / gemini-3-flash-preview	1.000	5/5 (1.000)	5/5 (0.783)	0.928	-
urgent-deadline-reminder	A	gemini / gemini-3.1-flash-lite-preview	gemini / gemini-3-flash-preview	1.000	5/5 (1.000)	5/5 (1.000)	1.000	-
urgent-deadline-reminder	B	gemini / gemini-3.1-flash-lite-preview	gemini / gemini-3-flash-preview	1.000	5/5 (1.000)	5/5 (0.917)	0.972	-
empathetic-support-reply	A	gemini / gemini-3.1-flash-lite-preview	gemini / gemini-3-flash-preview	1.000	5/5 (1.000)	5/5 (0.875)	0.958	-
empathetic-support-reply	B	gemini / gemini-3.1-flash-lite-preview	gemini / gemini-3-flash-preview	1.000	5/5 (1.000)	5/5 (0.800)	0.933	-
sales-outreach	A	gemini / gemini-3.1-flash-lite-preview	gemini / gemini-3-flash-preview	1.000	5/5 (1.000)	5/5 (0.842)	0.947	-
sales-outreach	B	gemini / gemini-3.1-flash-lite-preview	gemini / gemini-3-flash-preview	1.000	5/5 (1.000)	5/5 (0.800)	0.933	-
project-status-update	A	gemini / gemini-3.1-flash-lite-preview	gemini / gemini-3-flash-preview	1.000	5/5 (1.000)	5/5 (0.875)	0.958	-
project-status-update	B	gemini / gemini-3.1-flash-lite-preview	gemini / gemini-3-flash-preview	1.000	5/5 (1.000)	5/5 (0.783)	0.928	-
meeting-reschedule	A	gemini / gemini-3.1-flash-lite-preview	gemini / gemini-3-flash-preview	1.000	5/5 (1.000)	5/5 (0.875)	0.958	-
meeting-reschedule	B	gemini / gemini-3.1-flash-lite-preview	gemini / gemini-3-flash-preview	1.000	5/5 (1.000)	5/5 (0.800)	0.933	-
invoice-follow-up	A	gemini / gemini-3.1-flash-lite-preview	gemini / gemini-3-flash-preview	1.000	5/5 (1.000)	5/5 (0.875)	0.958	-
invoice-follow-up	B	gemini / gemini-3.1-flash-lite-preview	gemini / gemini-3-flash-preview	1.000	5/5 (1.000)	5/5 (0.800)	0.933	-
interview-invitation	A	gemini / gemini-3.1-flash-lite-preview	gemini / gemini-3-flash-preview	1.000	5/5 (1.000)	5/5 (0.958)	0.986	-
interview-invitation	B	gemini / gemini-3.1-flash-lite-preview	gemini / gemini-3-flash-preview	1.000	5/5 (1.000)	5/5 (0.750)	0.917	-

Comparative Analysis

Which strategy performed better?

Strategy A performed better with an overall average of 0.971, compared with 0.934 for Strategy B.

Biggest failure mode of the lower-performing strategy

The largest measured weakness of Strategy B was Professional Quality, its lowest average metric at 0.803. Within that hybrid metric, concision was the weakest automated component at 0.050.

Production recommendation

Recommend Strategy A for production because it achieved the stronger aggregate result under the same scenarios, model, judge, and metric definitions. Retain monitoring for fact fidelity and review the recommendation whenever the prompt or model changes.

Notes and Limitations

LLM-as-a-judge scores may vary despite a temperature of 0.0.

Free-tier availability and rate limits can change.

Mixed-provider runs are not fully controlled comparisons.

Human review remains necessary for sensitive or high-impact emails.